Optimization Project 1 – Group 29

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GitHub Repo: https://github.com/afroditifragiadaki/nyc-childcare-optimization

Introduction

Task 1

For this project, we were introduced to the prevailing issue of childcare deserts – regions in which the demand for child care far exceeds the available resources or slots for child care – throughout NYC. To eliminate these childcare deserts, we have two main options. The first involves building entirely new facilities (small, medium and large) that have pre-defined maximum capacities for 0–5 year old child-care. The second, or alternative, involves expanding and increasing the number of slots for child-care available at our existing facilities.

Our goal is to determine the minimum total funding required for NYC to eliminate these child-care deserts at every ZIP code, formulated as a Mixed-Integer Linear Programme (MILP). In this project, we are presented with two different scenarios with regards to cost structure and constraints, with Task 1 being more optimistic but less realistic. We will therefore solve for the minimum total funding with respect to these different scenarios.

1 Parameter Estimation

1.1 EDA and Data Cleaning

Before defining and solving the optimization, we first cleaned and merged the data. Four zip-code level datasets were used: child population by age group, regulated childcare facility records with capacity by age group (infant, toddler, preschool, school-age, children), average household income, and parental employment rate.

When loading the population data, we observed 28 ZIP codes with zero total population. These were therefore remapped to their nearest residential ZIP code. When merging average income and employment rate data, we observed 179 unique ZIPs, leaving four (10279, 11249, 11430, and 11439) without income or employment entries. After research: ZIPs 10279 and 11439 were dropped as non-residential regions with no usable data. ZIP 11430 covers a small population in and around JFK Airport, so its population data was transferred to the nearest residential ZIP. ZIP 11249 was identified as a gentrified, high-income area in Williamsburg, and given its demographic profile differed sharply from adjacent neighbourhoods, values were imputed from direct research rather than assigned from neighbours. After these adjustments, every ZIP has a unique, non-null entry for average income and parental employment rate. One facility within the desert ZIPs also reported zero total capacity and was dropped.

1.2 Parameters

The following ZIP-level quantities are computed from the cleaned data and used throughout the model:

Table 1: Parameters

Parameter	Description	Computation
P_z^{0-5}	Population aged 0–5 in ZIP z	From population data
P_z^{0-12}	Population aged 2 wks – 12 yr in ZIP z	$P_z^{0-5} + P_z^{6-12}$ from population data
S_z	Total existing childcare slots in ZIP z	Sum of total capacity across all facilities in ZIP z
S_z^{0-5}	Existing under-5 slots in ZIP z	Sum of infant + toddler + preschool capacity across all facilities in ZIP
$\underline{I}_z$	Average household income in ZIP z	From income data
e_z	Parental employment rate in ZIP z	From employment data
n_f	Current total capacity of facility f	From facility data

2 Desert Identification and Slot Deficits

2.1 Childcare Desert Identification

Based on the quantities defined above, we classified regions as being a childcare desert or not. This required first extracting the child population groups P_z^{0-5}, P_z^{6-12} for each ZIP, then aggregating the total slots S_z and under-5 slots S_z^{0-5} across all facilities, and then merging these with income and employment data. From this merged dataset, we were able to categorise each region by demand and determine whether it is a childcare desert.

Demand Classification

A ZIP is classified as a childcare desert depending on whether the area is considered **High-Demand** or **Normal-Demand**. High-demand areas have at least 60% parental employment or an average annual income of less than \$60,000:

$$\text{High-Demand}(z) = \mathbf{1}\big[e_z \geq 0.60 \;\text{ or }\; \underline{I}_z < \$60{,}000\big] \tag{1}$$

Desert Conditions

A ZIP with $P_z^{0-12} > 0$ is classified as a childcare desert if it fails either of the following two conditions.

Condition 1 – Demand-adjusted total capacity. Define the demand threshold τ_z as:

$$\tau_z = \begin{cases} \frac{1}{2} & \text{if High-Demand}(z) = 1 \\ \frac{1}{3} & \text{otherwise} \end{cases} \tag{2}$$

A ZIP is a childcare desert under Condition 1 if:

$$S_z < \tau_z \cdot P_z^{0-12} \tag{3}$$

Condition 2 – NYC Under-5 Policy. NYC also has a strict policy in which a region must have at least as many available childcare slots for children under the age of 5 as two-thirds of that population:

$$S_z^{0-5} < \frac{2}{3} P_z^{0-5} \tag{4}$$

Applying these rules yields **179 desert ZIPs**. All of the zip codes are desserts which is a truly concerning insight (except 1 that had 0 population of kids).

2.2 Computing Slot Deficits

For each desert ZIP we define two thresholds based on the conditions above. The total slot deficit and under-5 slot deficit are:

$$D_z^{\text{tot}} = \max\Big(0,\ \big[\tau_z \cdot P_z^{0-12} - S_z\big]\Big) \tag{5}$$

$$D_z^{0-5} = \max\Big(0,\ \big[\tfrac{2}{3} P_z^{0-5} - S_z^{0-5}\big]\Big) \tag{6}$$

A deficit of zero on one metric means that particular condition was already satisfied — the ZIP is still a desert due to failing the other condition.

3 Modelling (Task 1)

We now formulate the optimization model that minimises the total funding required to eliminate all childcare deserts in every ZIP code.

Index Sets

- Z: Desert ZIP codes, $|Z| = 179$.
- F_z: Existing facilities in ZIP z.

3.1 Decision Variables

Expansion slots at existing facilities:

$$x_f^{0-5} \in \mathbb{Z}_{\geq 0}, \quad x_f^{5-12} \in \mathbb{Z}_{\geq 0}, \qquad \forall\, f \in F_z,\ z \in Z$$

where x_f^{0-5} and x_f^{5-12} are the additional slots added for ages 0–5 and 5–12 respectively at existing facility f.

New facility construction:

$$y_z^S,\ y_z^M,\ y_z^L \in \mathbb{Z}_{\geq 0}, \qquad \forall\, z \in Z$$

Counts of new Small (100-slot), Medium (200-slot), and Large (400-slot) facilities to build in ZIP z. These facilities also have maximum under-5 capacities of 50, 100 and 200 respectively.

New build age-group allocation:

$$n_z^{0-5},\ n_z^{5-12} \in \mathbb{Z}_{\geq 0}, \qquad \forall\, z \in Z$$

Slots in newly constructed facilities allocated to ages 0–5 and 5–12 respectively.

Piecewise linearisation auxiliaries:

$$w_f \in \{0,1\}, \quad \delta_1^f,\ \delta_2^f \in \mathbb{R}_{\geq 0}, \qquad \forall\, f \in F$$

w_f selects the cost tier (0 = cheap, 1 = expensive); δ_1^f and δ_2^f are the slots allocated to each tier (more will be explained in the cost expansion formulation below)

3.2 Objective Function

$$\min\ Z = C_{\text{exp}} + C_{\text{build}} + C_{\text{equip}} \tag{7}$$

A. Expansion cost. The per-slot cost rates are:

$$c_1 = \frac{10{,}000 + 200\, n_f}{n_f}, \qquad c_2 = \frac{20{,}000 + 200\, n_f}{n_f} \tag{8}$$

Let $x_f = x_f^{0-5} + x_f^{5-12}$, $\delta_1^f + \delta_2^f = x_f$, and $M_f = \min(1.2\, n_f,\ 500)$. The tier bounds enforced by binary w_f are:

$$0 \leq \delta_1^f \leq n_f\,(1 - w_f), \qquad \forall\, f \in F \tag{9}$$

$$n_f\, w_f \leq \delta_2^f \leq M_f\, w_f, \qquad \forall\, f \in F \tag{10}$$

The total expansion cost is:

$$C_{\text{exp}} = \sum_{z \in Z} \sum_{f \in F_z} \left(\frac{10{,}000 + 200\, n_f}{n_f}\, \delta_1^f + \frac{20{,}000 + 200\, n_f}{n_f}\, \delta_2^f \right) \tag{11}$$

Note: We set c1 = (10,000 + 200nf)/nf, halving the fixed baseline from \$20,000 to \$10,000. This preserves the same structural form as c2, guarantees c1 is less than c2 for all nf ¿ 0, and remains decreasing in nf.

B. New construction cost.

$$C_{\text{build}} = \sum_{z \in Z} \left(\$65{,}000\ y_z^S + \$95{,}000\ y_z^M + \$115{,}000\ y_z^L \right) \tag{12}$$

C. Equipment surcharge. A flat \$100 surcharge applies to every slot newly allocated to children aged 0–5:

$$C_{\text{equip}} = \$100 \sum_{z \in Z} \left(\sum_{f \in F_z} x_f^{0-5} + n_z^{0-5} \right) \tag{13}$$

3.3 Constraints

$$\text{(C1)} \quad \sum_{f \in F_z} \left(x_f^{0-5} + x_f^{5-12} \right) + 100\, y_z^S + 200\, y_z^M + 400\, y_z^L \;\geq\; D_z^{\text{tot}}, \quad \forall\, z \in Z \tag{14}$$

$$\text{(C2)} \quad \sum_{f \in F_z} x_f^{0-5} + n_z^{0-5} \;\geq\; D_z^{0-5}, \quad \forall\, z \in Z \tag{15}$$

$$\text{(C3)} \quad n_z^{0-5} + n_z^{5-12} \;\leq\; 100\, y_z^S + 200\, y_z^M + 400\, y_z^L, \quad \forall\, z \in Z \tag{16}$$

$$\text{(C4)} \quad n_z^{0-5} \;\leq\; 50\, y_z^S + 100\, y_z^M + 200\, y_z^L, \quad \forall\, z \in Z \tag{17}$$

$$\text{(C5)} \quad x_f^{0-5} + x_f^{5-12} \;\leq\; \min(1.2\, n_f,\ 500), \quad \forall\, f \in F \tag{18}$$

Modelling Assumptions: Each ZIP code is self-contained: a child must reside in the same ZIP as their facility. All existing facilities are also assumed eligible for expansion regardless of their recorded status (e.g. Suspended), on the basis that the model signals where capacity is needed and suspended facilities can be reactivated accordingly.

3.4 Results (Task 1)

Gurobi solves the MILP to certified global optimality.

```
...  ========================================================
                OPTIMAL SOLUTION FOUND
     ========================================================

       Total Minimum Cost                  : $ 166,924,983.32
       ├─ Expansion Cost                   : $  62,116,283.32
       ├─ Construction Cost                : $  70,175,000.00
       └─ Equipment Surcharge              : $  34,633,700.00

       Expansion slots added (0-5)         :    224,675
       Expansion slots added (5-12)        :          3
         ├─ facilities on cheap tier (<nf)  :      7,433
         └─ facilities on exp   tier (≥nf) :        306
       New facilities built                :        611
         ├─ Small  (100 slots)              :          1
         ├─ Medium (200 slots)              :          2
         └─ Large  (400 slots)              :        608
       New build slots (0-5)               :    121,662
       New build slots (5-12)              :          0
```

Figure 1: Task 1 minimum cost breakdown by component.

The solution is shaped by two overarching forces. First, expansion is preferred over construction as cheap-tier expansion costs less per slot than new construction. Thus, the model exhausts available expansion capacity across existing facilities before turning to new builds. Second, the NYC Under-5 Policy is the dominant binding constraint – failing in all 180 eligible ZIPs compared to only 162 for the total capacity condition. As a result, almost all new capacity is directed toward 0–5 slots (346,337 of 346,340 total), and the $100 equipment surcharge on under-5 placements consequently accounts for a significant 20.8% of total cost.

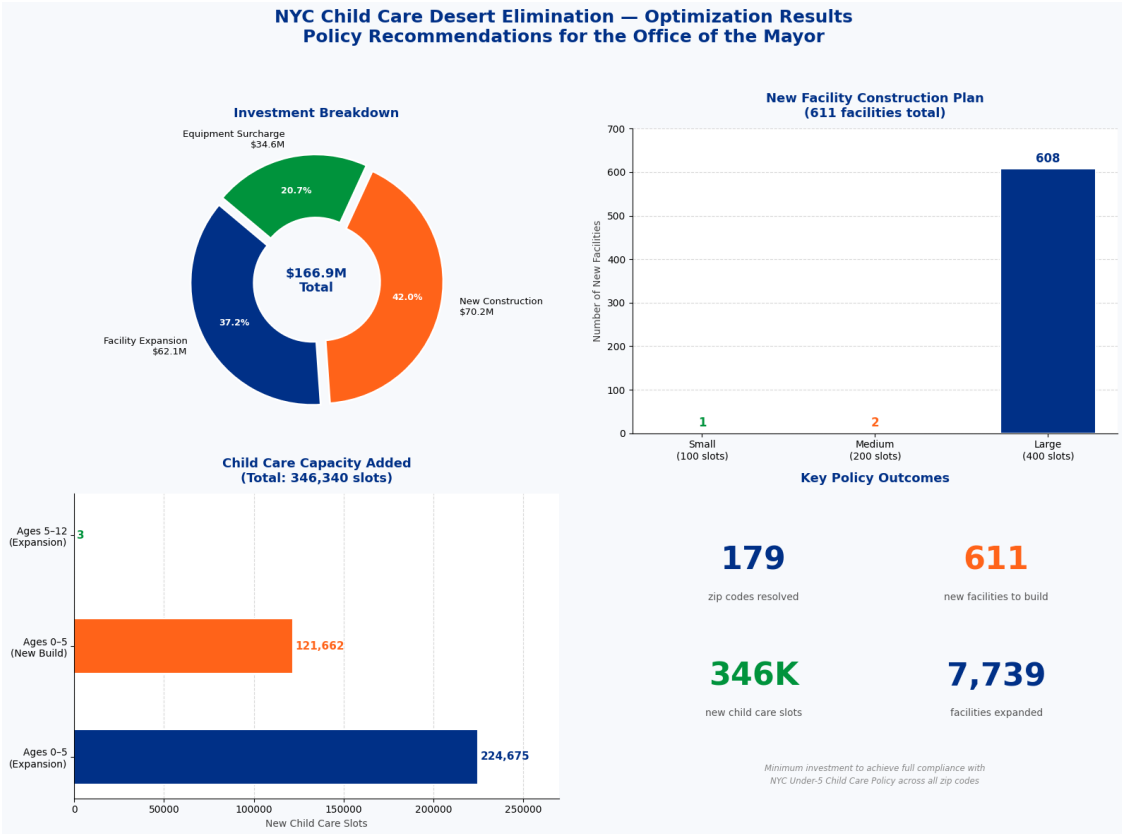

Figure 2: Recommended visualisation for Task 1

4 Task 2

Building on Task 1, Task 2 introduces a more realistic scenario with three key modifications: a **three-tier piecewise marginal expansion cost structure**, a strict **20% hard cap** on expansion at existing facilities, and a **minimum distance requirement** of 0.06 miles between any two facilities — existing or new — within the same ZIP code. New facility placement is also now restricted to specific candidate locations rather than anywhere in the city.

4.1 Parameter Estimation – EDA

The `potential_locations.csv` dataset contained 31,100 candidate building sites across 311 unique ZIP codes. Since only 180 matched the residential ZIP codes used in the model, 131 ZIPs required remapping: 29 via a manual lookup table of known commercial or PO-box ZIPs (e.g. 10020 → 10036 for Rockefeller Center), and the remaining 102 by assigning each to the nearest residential ZIP by Haversine great-circle distance. Separately, 169 of the 7,740 existing childcare facilities were missing latitude/longitude coordinates, which are required for the 0.06-mile distance constraint. These were filled via the Google Maps Geocoding API, querying each facility by name and city. After this step all facilities had valid coordinates.

4.2 Modelling

Distance Pre-processing

Before building the model, we pre-compute spatial feasibility using the Haversine formula to measure great-circle distances between locations. A minimum separation of 0.06 miles is enforced between any two facilities in the same ZIP code. This yields two outputs:

- **Infeasible locations:** any candidate site within 0.06 miles of an existing facility is ruled out entirely and removed from the decision space before the model is built.
- **Forbidden pairs:** any two candidate sites within 0.06 miles of each other form a forbidden pair – at most one of them may be selected. These are passed directly as constraints into the model.

After preprocessing, **3,135** of the 31,100 candidate locations are ruled out as infeasible, leaving **27,965** feasible sites. A further **87,011** forbidden pairs are identified among the remaining candidates.

Index Sets

Z and F_z: same as Task 1.

- L_z: Feasible candidate locations in ZIP z (after distance pre-processing).
- $\mathcal{F}$: Set of forbidden location pairs $\{(i,j) : \text{dist}(i,j) < 0.06 \text{ miles, both feasible}\}$.

4.3 Decision Variables

Expansion slots: x_f^{0-5} and x_f^{5-12} are defined identically to Task 1, with the only change being a tighter hard cap:

$$x_f = x_f^{0-5} + x_f^{5-12} \leq 0.20\, n_f$$

New facility construction (location-specific):

$$b_{l,S},\ b_{l,M},\ b_{l,L} \in \{0,1\}, \qquad \forall\, l \in L_z,\ z \in Z$$

$b_{l,s} = 1$ if a facility of size $s \in \{S, M, L\}$ is built at candidate location l. At most one size may be selected per location.

Piecewise linearisation auxiliaries:

$$t_{1,f},\ t_{2,f},\ t_{3,f} \in \{0,1\}, \quad \delta_{1,f},\ \delta_{2,f},\ \delta_{3,f} \in \mathbb{R}_{\geq 0}, \qquad \forall\, f \in F$$

$t_{k,f}$ selects which pricing tier is active; $\delta_{k,f}$ captures the total expansion x_f when tier k is active. At most one tier is active: $t_{1,f} + t_{2,f} + t_{3,f} \leq 1$.

4.4 Objective Function

$$\min\ Z = C_{\text{exp}} + C_{\text{build}} + C_{\text{equip}} \tag{19}$$

A. Expansion cost. Task 2 replaces the two-tier step function with a **three-tier non-continuous piecewise cost**. The entire expansion x_f is priced at the rate of whichever tier it falls into. Defining $b_1 = 0.10\,n_f$, $b_2 = 0.15\,n_f$, $b_3 = 0.20\,n_f$:

Table 2: Task 2 expansion cost tiers

Tier	Expansion range	Cost rate (applied to all slots)
1	$0 \le x_f \le b_1$	$(20{,}000 + 200\,n_f)/n_f$
2	$b_1 < x_f \le b_2$	$(20{,}000 + 400\,n_f)/n_f$
3	$b_2 < x_f \le b_3$	$(20{,}000 + 1{,}000\,n_f)/n_f$

Setting $x_f = \delta_{1,f} + \delta_{2,f} + \delta_{3,f}$, the tier bounds are:

$$0 \le \delta_{1,f} \le b_1 \cdot t_{1,f}, \qquad \forall\, f \in F \tag{20}$$

$$b_1 \cdot t_{2,f} \le \delta_{2,f} \le b_2 \cdot t_{2,f}, \qquad \forall\, f \in F \tag{21}$$

$$b_2 \cdot t_{3,f} \le \delta_{3,f} \le b_3 \cdot t_{3,f}, \qquad \forall\, f \in F \tag{22}$$

The expansion cost is:

$$C_{\text{exp}} = \sum_f \left[\frac{20{,}000 + 200\,n_f}{n_f}\,\delta_{1,f} + \frac{20{,}000 + 400\,n_f}{n_f}\,\delta_{2,f} + \frac{20{,}000 + 1{,}000\,n_f}{n_f}\,\delta_{3,f} \right] \tag{23}$$

B. Construction cost C_{build} and equipment surcharge C_{equip}: same as Task 1 (see Equations 12–13).

4.5 Constraints

(C1)–(C4) Total slot coverage, under-5 coverage, new build slot accounting, and new build under-5 cap are structurally identical to Task 1; the only notational difference is that ZIP-level counts $y_z^{S/M/L}$ are replaced by location-specific sums $\sum_{l \in L_z} b_{l,s}$.

$$\text{(C5)} \quad b_{l,S} + b_{l,M} + b_{l,L} \le 1, \quad \forall\, l \tag{24}$$

$$\text{(C6)} \quad \sum_s b_{i,s} + \sum_s b_{j,s} \le 1, \quad \forall\, (i,j) \in \mathcal{F} \tag{25}$$

$$\text{(C7a)} \quad t_{1,f} + t_{2,f} + t_{3,f} \le 1, \quad \forall\, f \in F \tag{26}$$

$$\text{(C7b)} \quad x_f = \delta_{1,f} + \delta_{2,f} + \delta_{3,f}, \quad \forall\, f \in F \tag{27}$$

$$\text{(C8)} \quad x_f^{0-5} + x_f^{5-12} \le 0.20\,n_f, \quad \forall\, f \in F \tag{28}$$

Modelling Assumptions (additional to Task 1): The distance constraint applies within each ZIP code only; no cross-ZIP distance checks are enforced. Existing facilities already within 0.06 miles of each other are grandfathered in; the constraint only governs new builds relative to other new builds and existing facilities. New facilities built on newly constructed sites are not eligible for expansion in this plan.

4.6 Results (Task 2)

```
============================================================
              BEST FEASIBLE (gap 0.0132%)
============================================================

  MINIMUM FUNDING REQUIRED (Realistic Scenario)
  $231,379,428.34

  Cost breakdown:
  ├─ Expansion Cost        : $   6,715,728.34
  ├─ Construction Cost     : $ 190,030,000.00
  └─ Equipment Surcharge   : $  34,633,700.00

  Solution details:
  Expansion slots (0-5)   :     18,198
  Expansion slots (5-12)  :          0
  New facilities built    :      1,660
    ├─ Small  (100 slots) :         13
    ├─ Medium (200 slots) :         11
    └─ Large  (400 slots) :      1,636

============================================================
```

Figure 3: Task 2 minimum cost breakdown by component.

Table 3: Comparison of Task 1 and Task 2 results

Metric	Task 1	Task 2	Change
Total minimum cost	\$166,924,983	\$231,444,428	+38.7%
Expansion cost	\$70,175,000	\$6,715,728	−89.2%
Construction cost	\$61,116,283	\$190,095,000	+170.9%
Equipment surcharge	\$34,633,700	\$34,633,700	0%
Expansion slots added	224,678	18,198	−91.9%
New facilities built	611	1,661	+171.7%

Task 2 increases total funding by approximately \$64.5 million relative to Task 1. This is caused entirely by the shift from expansion to construction: Task 1 added 224,678 expansion slots while Task 2 added only 18,198, requiring roughly 2.7× more new facilities. The distance constraint further limits where new facilities may be placed, adding logistical cost. Together, these constraints showcase the significant impact of realistic operational restrictions on childcare desert budgeting.

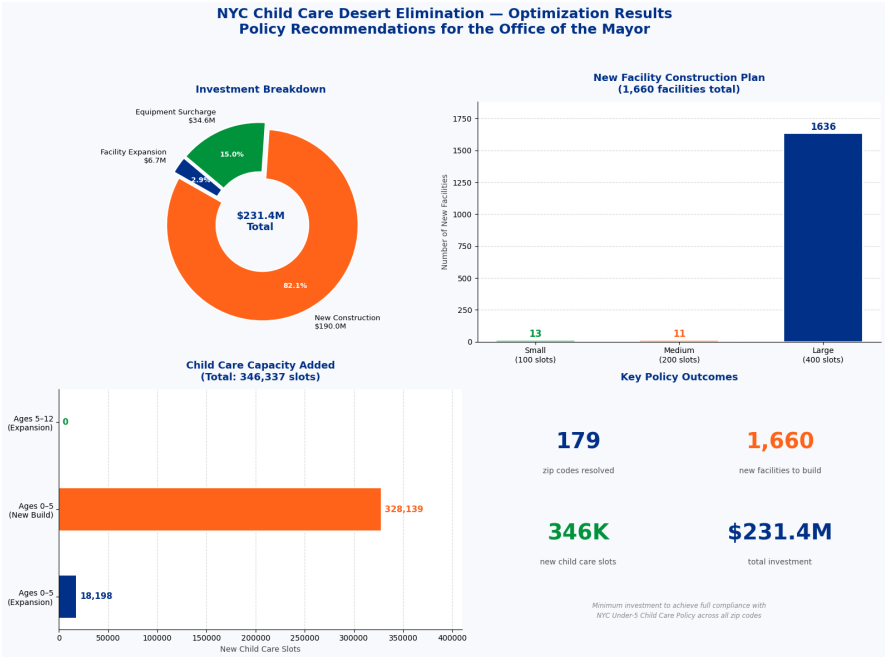

Figure 4: Recommended visualisation for Task 2

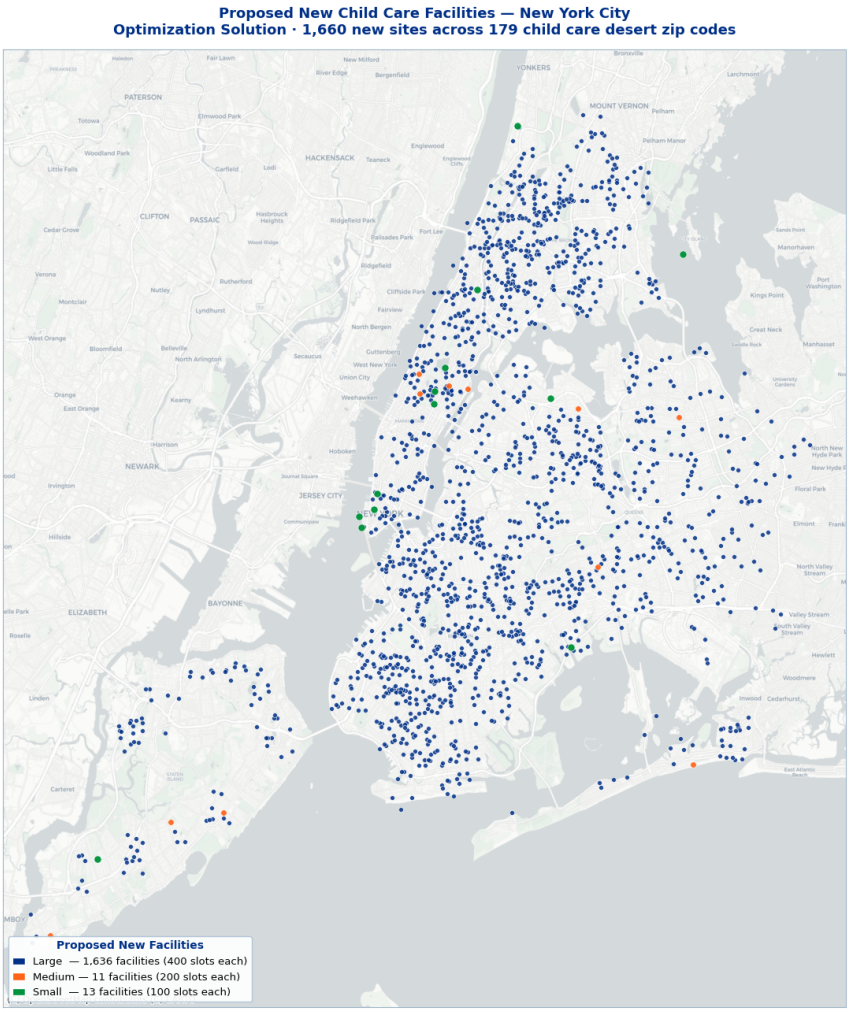

Figure 5: Recommended visualisation for Task 2